Slow-Fast Imitation Policy with Passive Variable Impedance Control for Tight Clearance Pin Insertion

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Abstract: Tight-tolerance robotic insertion requires transitions from free motion to contact. Tiny alignment errors may generate large force spikes and oscillations, resulting in jamming or damage. We hypothesize that purely reactive policies cannot achieve desirable contact properties, even using force feedback. Pin insertion on an industrial task with non-trivial geometry, $0.2\,\mathrm{mm}$ clearance and steel parts is studied to quantify this effect. We establish two diffusion policy baselines, with and without force observations, finding that both exhibit significant oscillations and large contact forces. This suggests that force observations alone are insufficient to provide smooth contact. Motivated by this finding, we develop one purely reactive and one predictive method for comparison. The reactive method uses an energy-tank based passivity law on the low-level controller during baseline rollouts. We further propose a hybrid architecture, that learns a separate high-frequency force-control policy and a low-frequency pose-control policy on kinesthetic demonstrations, together with a contact predictor. Both action modalities are then combined through a stiffness regulator and the contact prediction to produce desired behaviors.
We demonstrate that the energy-tank approach reduces peak contact forces by up to 32.8%, sustained forces by 33.5% and increasing signal to noise ratio (SNR) by up to 163% on the reference task. The predictive hybrid policy achieves a further 43% reduction in peak contact force, a 56.7% reduction in sustained forces and a 134% increase in SNR at the same success rate, confirming contact anticipations and high-frequency variable impedance are the key ingredient for safe, reliable tight-tolerance insertion.

Keywords: imitation learning, variable impedance control, diffusion policy, passivity, robotic insertion

1 Introduction

Insertion tasks are among the most common operations in manufacturing, yet tight-tolerance insertion remains a significant challenge for autonomous systems. When clearances shrink to fractions of a millimeter, small pose errors generate large contact forces that risk jamming or damaging the workpiece. This effect is strongly mediated by material properties: for compliant or plastic parts the contact dynamics are forgiving, but steel-on-steel contact leaves the controller almost no margin for recovery. In contrast to other contact-rich tasks like wiping, handling doorhandles, pushing etc., steel-on-steel pin insertion poses the following additional challenges, like discontinuous dynamics changes at the contact boundary, and multiple possible contact points [1].
Traditionally this is solved by creating extremely accurate initial alignment, through advanced vision techniques [2], appropriate hardware, as well as rigid and static fixtures, that allow for repeatable movement. The subsequent perfect knowledge of the task geometries can then be followed up with extensive controller tuning to arrive at the desired end-result.

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In this work we will treat the case of tight-tolerance steel pin insertion with uncertainty, warranting learning based methods to breach the gap. In practical terms we assume that the pose of the insertion geometry is partially unknown a priori, the detection of the hole is noisy and the low-level control law is static and generic (i.e. no pre-tuned stiffness profiles), while the given hardware does not achieve sub-millimetric accuracy reliably, as is common with generic cobots. This means, the applied policy needs to handle the discontinuous dynamics, change, fast force transients, and bridge the initial alignment and control uncertainties.

Imitation learning for contact-rich manipulation

Behaviour-cloning methods have matured rapidly over the past few years. Action chunking with temporal ensembling [3] and the diffusion policy framework [4] have become standard baselines, and contact-guided hierarchical representations [5] have further elevated success rates on dexterous tasks. Yet the benchmark tasks that drive this progress—wiping surfaces, rolling dough, and box pick-up [6]; opening doors and drawers and stacking blocks [4]; liquid transfer and container handling [7]; button pressing, tea pouring, and block stacking [8]—share a common property: contact is either light, spatially extended, or geometrically forgiving. None of these settings demands the millimetre-level alignment and simultaneous force regulation that characterise hard-material precision assembly. However, the toolbox for possibly solving such a task is expanding: While force/torque (f/t) feedback was often absent or treated as an auxiliary signal, recent work has begun to center this modality along with tactile sensing. FACTR [6] introduces a visual corruption curriculum that forces a transformer policy to attend to joint-torque readings rather than defaulting to vision, demonstrating a 43% improvement in generalisation to unseen objects on tasks such as gear assembly and box flipping. FoAR [9] integrates high-frequency f/t sensing with a future contact predictor to correct a visumotor policy purely in pose action space when a manually set contact probability threshold is detected. Aburub *et al.* [10] train a diffusion policy (DIPCOM) with that predicts a binary classified stiffness (high/low), in addition to the pose action. Shukla et al. [11] and Kang et al. [12] incorporate f/t observations into visuomotor diffusion policies for battery-cover fitting and compliant object prying, respectively, increasing success rates compared with pure visumotor baselines. TacDiffusion [13] shifts the action space entirely to the wrench domain, enabling impressive zero-shot transfer across sub-millimetre, at the cost of thousands of real-world demonstrations. VTLA [14] combines vision, tactile sensing, and language conditioning for insertion with impressive sim-to-real transfer, evaluated on plastic pegs with $0.6 - 2$ millimetres of clearnace. Force Generative Imitation Learning [15] learns to generate force commands from position trajectories via a model without memory, enabling stable feedback control for writing tasks; the approach is complementary but does not address the transient impact problem.

A consistent pattern across all of these works is that force sensing improves generalisation on moderately contact-rich scenarios; the specific challenge of suppressing *transient* impact forces on rigid parts at tight clearances has received comparatively little attention. Purely reactive policies might have a hard time mitigating this problem. RL-based approaches are currently the only class of methods that have been explicitly validated on hard-material, tight-clearance insertion with quantified force or contact metrics on a large scale. SERL [16] and HIL-SERL [17] achieve near-perfect success on PCB and RAM insertion on real robots, but require thousands of real-robot interactions and continuous human supervision throughout training. Sim-to-real approaches such as IndustReal [18], AutoMate [19], and FORGE [20] demand faithful contact simulation and were validated only on 3-D-printed plastic parts or on a steel peg inserted into a plastic hole. Zang et al. [21] reach 97%+ success on steel peg-in-hole at 0.1–0.2 mm clearance but require up to 4000 real-robot episodes per training phase. Across these systems the compliance strategy is either fixed or jointly optimised; the sample cost is substantial, and the infrastructure requirements are significant.

Other, explicitly model-based methods, like Contact-implicit MPC jointly optimise trajectories and contact forces by solving a non-convex, nonlinear complementarity problem [22]. Recent advances—consensus complementarity control (C3) [23], online residual learning [24], and complementarity-free formulations [25]—have improved real-time feasibility and reduced manual tuning. Nevertheless, the approach has not been demonstrated on tight-clearance steel insertion:

hole geometries are inherently non-convex, and the complementarity relaxations used for real-time tractability may not produce admissible contact forces under the stiff conditions imposed by metal-on-metal contact at the 0.1–0.5 mm scale.

An emerging alternative solution are world models to predict contact dynamics. Ideally, this reduces the need for real-robot interaction and has been popularized since the publication of Jepa [26]. DiWA [27] is trained on roughly four hours of unstructured play data, nad 50 demonstrations per skill on the Calvin benchmark [28]. Unified World Models [29] go further by jointly pre-training on large heterogeneous video and action datasets, with the world model serving as a bridge to action-free video. Even purely trajectory-based approaches such as policy-guided trajectory diffusion [30] and equality-constrained diffusion for direct trajectory optimisation [31] expand the toolbox for sample- efficient policy learning, but neither addresses the contact dynamics of rigid assemblies. OmniVTA [32] pairs a large-scale visuo-tactile dataset with a world-model-based framework that predicts short-horizon contact evolution and closes the loop at 60 Hz via a reflexive controller; it targets wiping and assembly at the level of friction changes, not sub-millimetre rigid insertion.

Research gap and contributions

Taken together, the literature reveals a gap in data-efficient imitation learning methods that specifically handle the case of steel-on-steel insertion with tight clearances. Further, contact metrics are hardly evaluated This omission is understandable for tasks where contact is light, but it needs to be specifically addressed in the scope of this work.

We present an empirical study demonstrating that naïve generative diffusion policies trained for steel-on-steel insertion—with and without force/torque feedback—fail to solve tight-clearance contact-rich steel insertion despite achieving better performance on equivalent plastic-part tasks, exposing the critical role of contact dynamics and material properties in policy robustness and revealing a contact-quality reporting gap in the existing imitation learning literature. To characterize this, we use peak forces, 95th-percentile forces, average sustained forces, and mean and standard deviations the signal to noise ratio (SNR) as key performance indicators. The force metrics cover worst-case and average efforts, while the SNR is a proxy for the intensity of oscillations. Building on these findings, we introduce an energy-tank-based reactive passivity controller that significantly reduces sustained and peak contact forces as well as force oscillations during policy rollouts, and further propose a predictive hybrid policy architecture that combines a learnt contact prediction with a high-frequency force control policy and a low-frequency visuomotor policy through a variable stiffness law with stability guarantees—outperforming both the naïve baseline and the reactive energy-tank controller in empirical validation.

2 Methods

Task Description and Experimental Setup

The task is a steel pin (rectangular head, threaded M8 shaft) insertion with tight tolerance ($0.2\,\mathrm{mm}$ radially) on a Franka FR3 robot arm. The target structure is a steel wheel with holes $8\,\mathrm{mm}$ diameter. It is held in place by MDF semi-rigid fixtures attached to the robot's mounting giving some play on contact. The task is considered for per hole (i.e. the insertion can happen anywhere on the reachable part of the wheel, but one task sequence is one hole insertion). Roll and pitch orientations are kept constant to focus on the contact quality. Figures 1 shows an experimental run with the pin geometry and a correctly seated insertion through the overlay. All observations and actions are expressed relative to this detected hole position, allowing the policy to generalise across any hole without retraining.

All proposed policies are trained on the same demonstration data (excluding the f/t information for the purely pose-based policy) with the same augmentation (including but not limited to, for image : color jiter, Gaussian noise and for proprioreceptive state: position, rotation and wrench noise) and preprocessing pipeline.

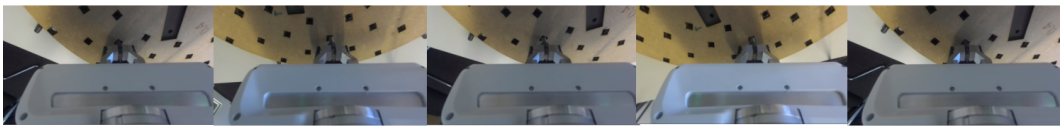


Figure 1: POV Timelapse of the insertion task.

Demonstrations are collected via kinesthetic teaching in gravity compensation, with high-frequency robot state logging (1 kHz), wrist RGB video (30 Hz) and wrist-mounted f/t-stream at over (200 Hz). All policies uses a ResNet-18 visual backbone with proprioceptive history. *Baseline 1* and *Baseline 2* output a diffusion-based action of end-effector poses, at 30 Hz. *Baseline 2* additionally uses f/t input. *Energy Tank* includes a separate energy-tank-based compliant control layer (see 2.1) updating controller stiffness at 200 Hz and using *Baseline 1* as reference action. *Hybrid* denotes our proposed architecture for predictive stiffness modulation (see 2.2). All pose targets are tracked by a low-level impedance controller (for *baseline 1* and baseline 2 the gains are fixed, see C.2) at 1 kHz. Training uses standard diffusion and optimization techniques, with full details deferred to Appendix B.

Table 1: Summary of evaluated architectures. F/T = force/torque sensor; K = Cartesian stiffness matrix.

Architecture	**Observations**	**Actions**
Baseline 1	RGB, proprio @ 30 Hz	Pose @ 30 Hz
Baseline 2	RGB, proprio, F/T @ 30 Hz	Pose @ 30 Hz
Energy Tank	RGB, proprio @ 30 Hz	Pose @ 30 Hz
Hybrid	RGB, proprio @ 30 Hz, F/T @ 200 Hz	Pose @ 30 Hz, K @ 200 Hz

2.1 Energy-Based Variable Stiffness Controller

For tight-tolerance steel insertion, no fixed stiffness satisfies both the free-space tracking and the contact force requirements. Thus, we introduce an alternative approach based on energy-tanks [33], that is purely reactive. Thereby the translational stiffness K of the low-level impedance controller is modulated in real time, based on the contact power. Note that the (instantaneous discrete-time) contact power is negative if velocity and force are in opposite directions, i.e. moving against a wall at timestep k, and positive (or zero) when moving into free space. [colgate]

$$P_{\text{env}}(k) = f_{\text{ext}}(k) \cdot \dot{x}(k), \tag{1}$$

smoothed with an exponential moving average ($\alpha_{\text{ema}} = 0.05$). The asymmetric stiffness modulation law is:

$$\dot{K} = \begin{cases} \alpha\, \bar{P}_{\text{env}} & \text{if } \bar{P}_{\text{env}} < 0 \quad \text{(soften)} \\ \beta\, (K_{\max} - K) & \text{if } \bar{P}_{\text{env}} \geq 0 \quad \text{(recover)} \end{cases} \tag{2}$$

When $\bar{P}_{\text{env}} < 0$, the environment is absorbing energy and K decreases, increasing compliance. When $\bar{P}_{\text{env}} \geq 0$, stiffness recovers toward $K_{\max}$. The softening gain $\alpha = 5 \times 10^5$ per axis is chosen to produce stiffness rates of the correct magnitude for the milliwatt-range contact power transfer during insertion; the recovery gain $\beta = 1$ ensures gradual stiffening and was manually tuned.

Passivity Bound on the Recovery Branch

The recovery branch is designed following the passivity conditions of Kronander and Billard [34]. Under this coupling, and following the discrete-time extension of Theorem 1 in [34], a sufficient condition for passivity during stiffness *increase* ($\dot{K} > 0$) is that the stiffness update satisfies

$$\Delta K \preceq \frac{2\,\alpha\, K(t)}{1 + \frac{\alpha\zeta}{2}\sqrt{K(t)\,\text{diag}(\Lambda(t))}}\,\Delta t, \tag{3}$$

where Δt is the discretization time step and $\alpha > 0$ is a scalar design parameter inherited from the continuous-time passivity condition. Thereby we used $D = \zeta\sqrt{K\,\mathrm{diag}(\Lambda)}$, with $\Lambda = (JM^{-1}J^{\top})^{-1}$ denoting the task-space apparent inertia projected along each Cartesian axis, and $\zeta > 0$ a constant damping gain.
The softening branch ($\dot{K} < 0$) is left unconstrained, as reductions in stiffness strictly decrease stored energy and therefore preserve passivity.

The parameter α is selected adaptively at each timestep as

$$\alpha(k) = \frac{1}{\lambda_{\max}(D^{-1/2} H D^{-1/2})} \tag{4}$$

where $\lambda_{\max}(D(t))$ denotes the maximum eigenvalue of the congruence transformation $(D^{-1/2}HD^{-1/2})$ and H denotes the desired inertia from the impedance law. In our case $H = \Lambda(k)$, the real operational space Inertia of the robot. This choice recovers the maximum admissible scaling consistent with the generalized eigenvalue condition underlying the stability result. Thereby, it is implicitly assumed that variations in $\Lambda(t)$ are negligible over the duration of contact. A detailed derivation can be found in 3

2.2 Hybrid Predictive Architecture

Our third approach, the main contribution, combines a slow/fast diffusion policy with contact prediction, similar to [35] and [9]. *Baseline 2*, alongside a contact predictor and a fast force policy. The force policy predicts 6D applied forces from the expert demonstrations and is trained on 200 Hz force and proprioceptive states. Importantly, the force policy is trained exclusively on data collected during contact phases, by sampling only state/force pairs that exceed a certain force threshold. The contact predictor is trained on proprioceptive state, and force observations at 30 Hz, outputting a contact probability in $[0, 1]$ using the same labels. However, instead of modifying the actions of the global policy directly, we use the force policy and the contact predictor to adapt the impedance parameters of the controller. The architecture is illustrated in Figure 5, yielding a similar structure as in [36]. Note that this is consistent with the separate training approach: By approximately visit the same states as we would using the visumotor policy alone, while recovering the distribution of measured forces by modulating impedance. If we change the pose action instead of the stiffness, the dynamics gap between the operator and controller would not be bridged, which might throw the force policy out of distribution (ood) and subsequently, by visiting different states, the pose policy as well.

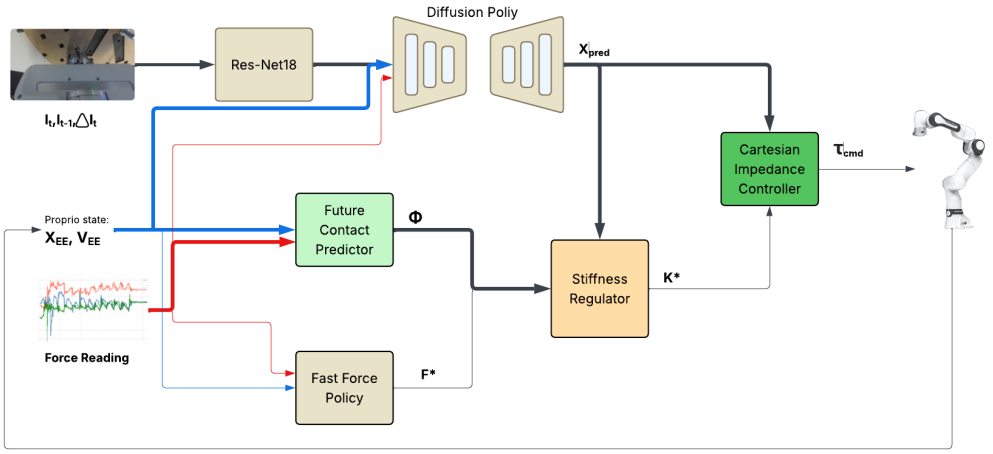


Figure 2: Proposed hybrid policy architecture. Thick arrows denote 30 Hz information streams, while slim ones denote 200 Hz or 1 kHz streams

We treat F^* is the reference force generated by the force policy, and that Δx denotes the difference between the global action at timestep k and $k+1$, i.e., the commanded displacement over one control step. In steady state, this implies that an approximate force $K\Delta x$ is applied at timestep k, which should match the desired force F^*.

We can therefore define a desired stiffness as

$$K^*(t) = \frac{F^*(t)}{\Delta x(t)}. \tag{5}$$

The stiffness is then updated using the contact prediction α.

$$K(t+1) = \big(1 - \phi(t+1)\big) * K_0(t) + \phi(t+1)K^*(t), \tag{6}$$

where K_0 is the nominal controller stiffness and K^* is the desired stiffness, consistent with the force policy. $\phi(k+1)$ is the contact probability of the next step given by the contact predictor. Then, the same passivity-preserving clipping bound as the energy-tank approach in equation C.3 is applied to ensure stability of the coupled impedance system.

3 Results

The main challenge is to seat the pin through two different contact phases. As the robot starts in free-space it might hit the MDF cutout before entering the cavity and only then can it find the actual hole. When the pin is seated (given by z-height threshold), the task is counted as success. Note that the main point of interest however is to get contact dynamics that are as smooth as possible.
For the subsequent comparisons we report the results for insertion tasks with steel-on-steel, and PLA-on-steel contacts for the previously introduced methods. Table 2 shows the average peak forces per run the average 95th-percentile contact Force magnitudes, the RMS (sustained) Forces in every direction and the median SNR and standard SNR deviation. Figure 3 shows the distribution of key metrics thorughout architectures for the steel-on-steel task. The reported SNR's are computed as $\frac{S_{0-1}}{N_{1-600}}$, where S_{0-1} is the sum of the frequency contributions between 0 and 1 Hz, while N_{1-600} is the sum of frequency contributions from 1 to 600 Hz. The frequency spectra are computed via FFT.

Table 2: Success rates and contact force metrics for the steel and PLA variants.

(a) Steel variant

Policy	Success	Max $\|F\|$ [N]	95th $\|F\|$ [N]	RMS F_x [N]	RMS F_y [N]	RMS F_z [N]	SNR [-]	SNR σ [-]
Baseline 1	18/20	76.75	44.59	11.96	10.43	15.93	7.1	11.2
Baseline 2	16/20	57.19	32.05	4.31	6.60	13.46	8.8	14.0
Energy Tank	18/20	38.37	21.82	6.93	6.30	8.95	23.2	28.7
Hybrid	19/20	21.86	10.62	3.46	4.53	3.87	54.4	95.6

(b) PLA variant

Policy	Success	Max $\|F\|$ [N]	95th $\|F\|$ [N]	RMS F_x [N]	RMS F_y [N]	RMS F_z [N]	SNR [-]	SNR σ [-]
Baseline 1	14/20	55.28	22.13	5.69	4.24	10.26	6.6	72.4
Baseline 2	13/20	83.87	38.94	4.52	5.73	16.72	5.3	11.7
Energy Tank	12/20	34.80	15.63	6.66	3.83	6.31	19.3	40.3
Hybrid	13/20	33.29	13.11	5.08	2.87	5.26	14.5	23.1

While incorporating force/torque observations in Baseline 2 yields a modest reduction in contact forces (25%), peak forces (25%), and mean signal-to-noise ratio (23.9%) relative to Baseline 1, this improvement does not constitute a qualitative difference in behaviour and is in fact similar with the performance gap observed between Baseline 1 evaluated on PLA versus steel pegs. The energy-tank controller combined with Baseline 1 achieves a more substantial reduction, decreasing average contact forces (in z) by approximately 33.5% and peak forces by 32.9% compared to Baseline 2; however, it continues to exhibit unreasonably high force peaks during insertion, suggesting that reactive compliance alone is insufficient for robust peg-in-hole manipulation. The proposed hybrid architecture, combining the predictive contact module with the impedance regulator, outperforms all

(c) Force Metric Distributions

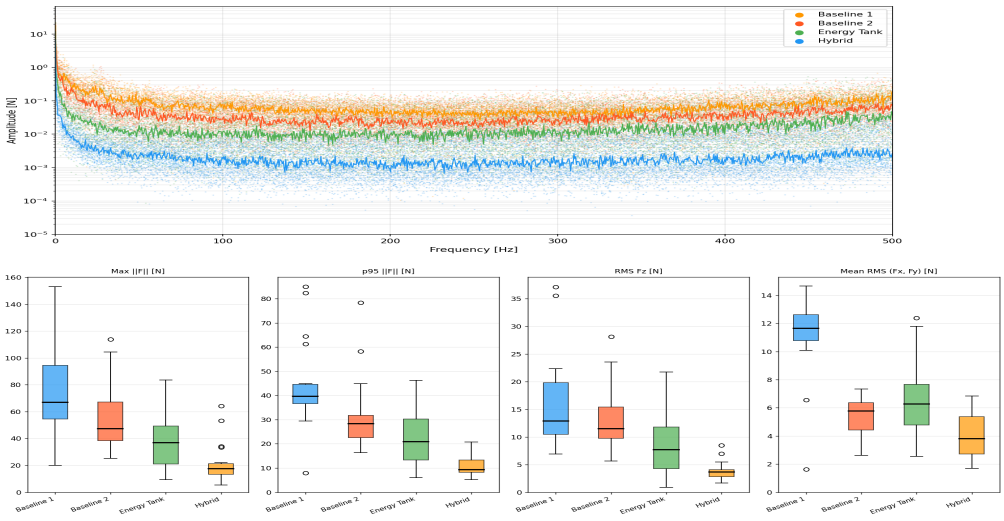


Figure 3: (c) Scatter plot of frequency domain distribution of force magnitude across all architectures and steel-on-steel task. Each run contributes one data-point per frequency bin. The continuous line shows the median. (d) Box plots of contact force distributions across all architectures and the steel task. Mean RMS (Fx, Fy) denotes the average radial forces acting on the pin.

other methods by a considerable margin. Compared to the energy-tank controller, it reduces peak forces by 43%, sustained contact forces by 56.75%, and increases mean signal-to-noise ratio by approximately 134%, as summarised in Table 2. Figure 3 illustrates the distribution of fourier spectra throughout the runs, and clearly shows that noisy vibrations for the *Hybrid* policy are magnitudes smaller than for *Energy Tank*, which are in turn magnitudes smaller than *Baseline 1&2*.

4 Discussion

We can start with an interesting observation regarding the differences in the PLA-on-steel (PLAoS) ans steel-on-steel (SoS) task. The success rate declines for all policies, which can be attributed to a changing vision modality and slightly different dynamics, leading to ood states. However, on the force metrics, the policies actually using f/t sensing experience a larger negative effect, which can be explained like this: Since *Baseline 1* does not know about forces, it just benefits from less aggressive contact dynamics; The same can be same for the energy tank, implying good generalization properties. The force-based policies however expect certain contact force distributions (with *Hybrid* even enforcing them as actions) that are not recovered in the different material combination. Still, the Hybrid policy outperforms the Baselines by a large and the *Energy Tank* by a small margin.

For the SoS task, Baseline 2 yielded a moderate, although highly insufficient improvement over the purely visuomotor Baseline 1. This further suggests that a pose-based policy cannot easily exploit raw F/T signals without an explicit mechanism to act on them or to balance their training with rich vision-data, which reflects findings as [37] and [6]. Additionally, higher frequency actions might be necessary to capitalize on force feedback [14] [35]. *Energy tank* reduced peak (33%) and sustained forces (61%), as well as oscillations relative to the fixed-stiffness baselines, achieving similar results in SoS and PLAoS. The *Hybrid* policy further demonstrated the benefit of predictive compliance and hybrid force/pose policies by reducing contact forces even further (50% peak force, 25% sustained force and 33% oscillation reduction). Crucially, the proposed policy makes no assumption about the underlying impedance parameters and can therefore be deployed on top of any nominal controller configuration without retuning, unlike e.g. [10] — a practically relevant

property when hardware setups vary across deployments.

5 Limitations

While the proposed policy architecture reduces contact forces and oscillations by considerable margins, it fails to generalize to the PLA task, as can be seen by the reduced success rate. This indicates a high sensitivity to the force readings, which is a two-way street: On one hand the contact behaviour for the trained task is very reactive, on the other hand the explicit force reference does not generalize to different dynamics (and ironically, it generalizes worse than *Baseline 1*, relatively speaking. While this may be acceptable for repeated trainings as the data requirements for the policy are low (20 demonstrations), better generalization properties could be achieved by encoding the environment dynamics, normalizing the force observations (such as looking at relative changes) or using techniques as positional embeddings or Fourier feature encodings.

On another note, the validation and training has only been done on a single task with considerable geometric simplifications (no roll and pitch actions and no explicit yaw goal). This indicates that for more general manipulation tasks, it may be necessary to increase the complexity of the system. Again the contact predictor would need to be increased in dimensionality, to account for the contact direction. In its current state the architecture could be expected to generalize well to other top-down insertion tasks that are dominated by top-down normal contact, but would struggle on tasks where there can be more than one dominant contact direction or significant orientational misalignment.

Another main failure mode comes from surface friction or jamming. Here the policy could be improved by adding recovery mechanisms [38], data that captures specific failure modes [39], explicit friction modelling as in CI-MPC [23] or online disturbance estimation.

Finally, as the policy assumes known hole locations, the success rate deteriorates drastically as the initial alignment error grows. For the presented validations, the initial alignment error is in the range of $0 - 10mm$, which is in line with similar policies [20]. For less structured tasks or harder perception challenges this might require another mitigation strategy.

On a minor note, the approach does not deliver stability guarantees for the full pipeline. Even if the high-frequency stiffness corrections by themselves keep the system stable, wrong contact predictions or artifacts in the force policy (such as abnormally large force predictions) may still drive the system into unstable states.

6 Conclusion

We present a unified force/pose architecture with contact prediction for tight-clearance pin insertion tasks and show that this predictive element is necessary to mitigate high contact forces when working with highly rigid (steel) objects. Empirical validation showed that the contact behaviour of a pure visumotor diffusion policy is poor, and that simply including f/t observations does not significantly change this fact. Additionally the proposed hybrid architecture with contact prediction outperformed both baselines, as well as a reactive impedance policy, based on the enregy-tank method. Finally, our policy is physically meaningful in the way it integrates with impedance control, providing a consistent formulation without any assumptions on the controller stiffness and without manual tuning at inference time.

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A Appendix

A.1 Policy Architectures

The policy receives four observation modalities. The 9-channel image input ($9 \times 224 \times 224$) is processed by a modified ResNet-18 backbone (pretrained on ImageNet) producing a 128-dimensional visual embedding. A proprioceptive history of 5 timesteps—end-effector pose (7D), velocity (6D), and force derivative (3D)—is processed by a two-layer MLP to produce a 64-dimensional embedding. The two embeddings are concatenated into a 192-dimensional observation vector that conditions the diffusion process.

The action space consists of absolute end-effector poses in the hole-centred frame (7 values per timestep: position + quaternion), forming a chunk of 64 future poses. The policy is a conditional diffusion model with a 1D U-Net denoising network. Training uses a cosine noise schedule over 100 diffusion steps; at inference, DDIM with 10 steps meets the 30 Hz control rate. Consecutive chunks overlap, and the executed action at each timestep is a weighted average across all overlapping predictions.

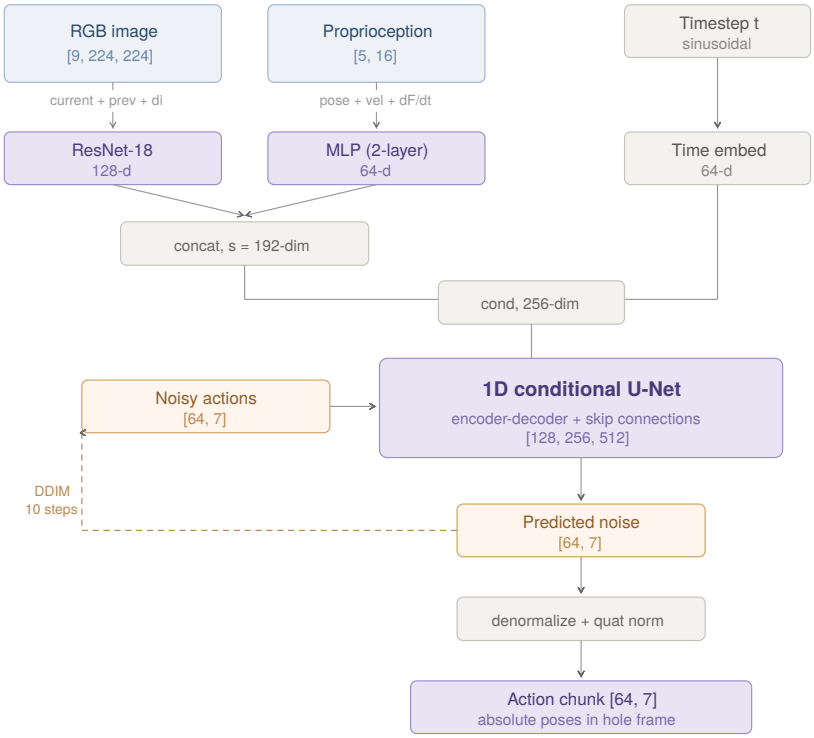


Figure 4: *Baseline 1* Diffusion policy architecture. The RGB image stack (9 channels) is encoded by a modified ResNet-18, and the proprioceptive history is encoded by a two-layer MLP. Their concatenation ($s \in \mathbb{R}^{192}$) is combined with a 64-dimensional sinusoidal time embedding to form the 256-dimensional conditioning vector. A 1D conditional U-Net predicts the noise component; at inference, DDIM runs 10 denoising steps to produce the 64-step action chunk.

B Training and Implementation Details

B.1 Data Collection and Preprocessing

Demonstrations are collected via kinesthetic teaching in gravity compensation mode. Robot state, including end-effector pose, joint states, and external forces, is recorded at 1 kHz, while wrist-mounted RGB video is recorded at 30 Hz. Each demonstration lasts approximately 10 seconds, and 20 demonstrations are collected per task variant.

Data is temporally aligned across modalities. Idle phases are automatically removed, and velocity and force derivatives are filtered. Visual observations and proprioceptive histories are constructed using temporally stacked frames and state histories.

B.2 Policy Architecture

The policy follows a conditional diffusion formulation. Visual inputs consist of a 9-channel image tensor of size $9 \times 224 \times 224$, processed by a modified ResNet-18 backbone pretrained on ImageNet, producing a 128-dimensional embedding. Proprioceptive history includes 5 timesteps of end-effector pose (7D) and velocity (6D), which are embedded jointly with the visual features.

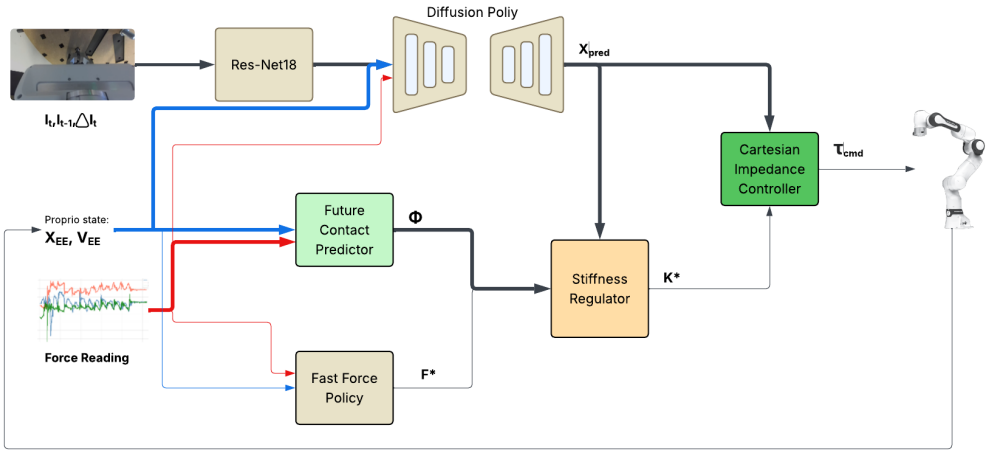


Figure 5: Proposed hybrid policy architecture. The RGB image stack (9 channels) is encoded by a modified ResNet-18, and the proprioceptive history is encoded by a two-layer MLP. Force Data is applies thorugh an MLP-diffusion network into a learnable embedding used for the fast policy. Their concatenation ($s \in \mathbb{R}^{192}$) is combined with a 64-dimensional sinusoidal time embedding to form the 256-dimensional conditioning vector. A 1D conditional U-Net predicts the noise component; at inference, DDIM runs 10 denoising steps to produce the 64-step action chunk.

The action space consists of absolute end-effector poses in the task frame (position and quaternion), predicted as a horizon of 64 future steps. A 1D U-Net is used as the diffusion denoiser. Training uses a cosine noise schedule over 100 diffusion steps.

C Inference, Control and Training

At inference time, DDIM sampling with 10 diffusion steps is used to meet the 30 Hz control rate. Consecutive action chunks overlap in time, and executed actions are obtained by weighted averaging over overlapping predictions.

The policy outputs are interpreted as desired end-effector pose targets.

C.1 Energy-Based Stiffness Adaptation

A separate energy-based module runs at 200 Hz, estimating contact-related quantities and updating task stiffness. The stiffness signal is sent independently of the pose policy.

C.2 Control Architecture

The low-level controller is a Cartesian impedance controller running at 1 kHz. It tracks the most recently received pose target and stiffness value, treating both as independent asynchronous command channels without requiring synchronization or shared state. The robot's end-effector is governed by a Cartesian impedance controller tracking a desired pose x_d with stiffness K and damping D:

$$f = K(x_d - x) + D(\dot{x}_d - \dot{x}). \tag{7}$$

And the commanded torques are sent to the robot as

$$\tau = J^T f + c(q, \dot{q}) + g(q), \tag{8}$$

where $c(q, \dot{q})$ and $g(q)$ denote coriolis and gravity torques respectively.

The nominal stiffness is set to

$$K = \mathrm{diag}(1500,\ 1500,\ 2000,\ 100,\ 100,\ 50), \tag{9}$$

and the damping matrix $D(k)$ is chosen as

$$D(k) = \zeta \cdot \mathrm{diag}\Big(\sqrt{K \cdot \Lambda(k)}\Big), \tag{10}$$

where $\sqrt{\cdot}$ denotes the element-wise square root, $\Lambda(k) \in \mathbb{R}^{6\times 6}$ is the Cartesian inertia matrix of the robot at time step k, and $\zeta = 2.3$ is a tunable damping ratio.

C.3 Training

The model is trained using AdamW with cosine annealing and exponential moving average weights. Contact-aware sample reweighting and multimodal augmentation are applied to both visual and proprioceptive inputs.

Derivation of the Maximum Scaling Factor α

To find the largest $\alpha > 0$ such that $\alpha H - D \preceq 0$, we proceed as follows:

1. Formulation of the Inequality

The condition for the matrix to be negative semi-definite is:

$$\alpha H - D \preceq 0 \implies \alpha H \preceq D \tag{11}$$

Dividing by α and defining $\lambda = \frac{1}{\alpha}$:

$$H \preceq \frac{1}{\alpha} D \implies H \preceq \lambda D \tag{12}$$

To maximize α, we must find the minimum λ that satisfies this inequality.

2. Congruence Transformation

Assuming D is a positive definite diagonal matrix, we perform a congruence transformation using $D^{-1/2}$:

$$D^{-1/2} H D^{-1/2} \preceq \lambda \left(D^{-1/2} D D^{-1/2}\right) \tag{13}$$

Since $D^{-1/2} D D^{-1/2} = I$, the expression simplifies to:

$$D^{-1/2} H D^{-1/2} \preceq \lambda I \tag{14}$$

3. Spectral Bound

Let $A = D^{-1/2} H D^{-1/2}$. The inequality $A \preceq \lambda I$ holds if and only if λ is greater than or equal to the maximum eigenvalue of A:

$$\lambda \geq \lambda_i(A) \quad \forall i \implies \lambda_{\min} = \lambda_{\max}(A) \tag{15}$$

4. Final Solution

Substituting $\lambda = \frac{1}{\alpha}$ back into the expression yields the largest allowable value for α:

$$\alpha_{\max} = \frac{1}{\lambda_{\max}(D^{-1/2} H D^{-1/2})} \tag{16}$$

Equivalently, this is the reciprocal of the maximum generalized eigenvalue of the pair (H, D):

$$\alpha_{\max} = \frac{1}{\lambda_{\max}(H, D)} \tag{17}$$